A Rough Multivariable Model of Energy Loss in Badminton Racket–Shuttlecock Collisions

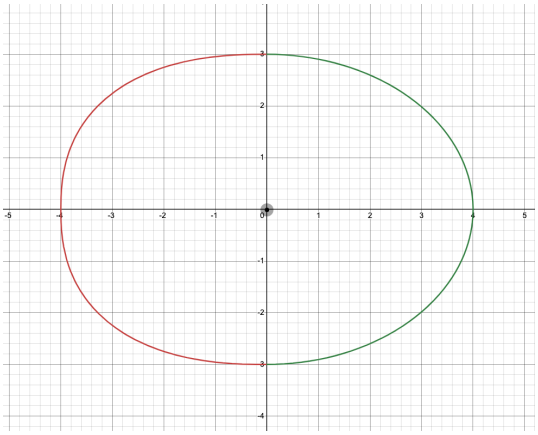
By Suraj Gupta

Abstract

This paper aims to model the energy loss of a shuttlecock during collision with a badminton racket using a multivariable function dependent on the impact location on the string bed and the string tension of the racket.

Shape of the Head

The shape of a badminton racket's head can be roughly represented using a hybrid ellipse-superellipse. For every $x \geq 0$, the curve is given by $\left(\frac{x}{4}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ (ellipse) and for every $x < 0$, the curve is given by $\left|\frac{x}{4}\right|^{2.4} + \left|\frac{y}{3}\right|^{2.4} = 1$ (superellipse).



[Desmos]

To maintain simplicity, this model ignores drag, string material, and string-bed physics.

The Parameters

The idea is to create a model that accurately matches experimental results based on the impact location of the shuttle on the string bed and the tension of the racket.

Calculating the percentage of energy after impact as compared to before impact and subtracting it from 100% gives the energy loss percentage.

Let L represent the loss %

$$L = 100\left(1 - \frac{E_v}{E_u}\right)$$

where E_v is energy after impact and E_u is energy before impact.

$$E_u = \frac{1}{2}mu^2, E_v = \frac{1}{2}mv^2$$

$$\Rightarrow L = 100\left(1 - \frac{{}^1\!/_2\, mv^2}{{}^1\!/_2\, mu^2}\right) = 100\left(1 - \frac{v^2}{u^2}\right)$$

Now, $v^2 = 2gh_v$ and $u^2 = 2gh_u$

$$\Rightarrow L = 100\left(1 - \frac{2gh_v}{2gh_u}\right) = 100\left(1 - \frac{h_v}{h_u}\right)$$

The impact location can be calculated through a location index by taking the arithmetic mean of the average distance of the point of connection from the points on the racket frame with the same abscissa and the distance of the impact location from the centre.

Let δ represent the impact location.

$$\delta = \frac{\left(\frac{|y(h) - k| + |y(h) + k|}{2} + \sqrt{h^2 + k^2}\right)}{2}$$

where (h, k) is an arbitrary point lying within the racket head, and

$$y(h) = 3\sqrt[t]{\left(1 - \left|\frac{h}{4}\right|^t\right)}, t = \begin{cases} 2 \ \forall x \geq 0 \\ 2.4 \ \forall x < 0 \end{cases}$$

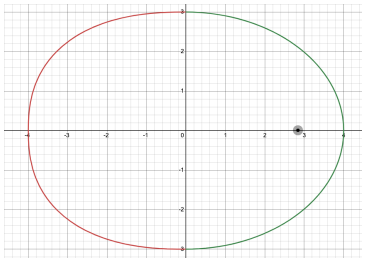
Experimental Results

Through an experimental setup, the rebound heights of the shuttle when dropped at different points on the rackets were considered. The experimental setup used three rackets, each with a different tension (the tension of the rackets is a rough approximation and not accurately measured)

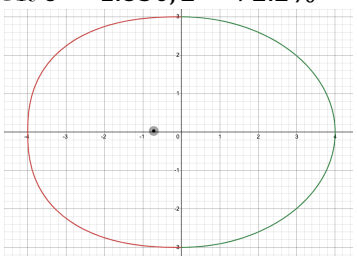
		
Li-Ning XP 901-PV SINDHU Signature Series	Yonex Nanoflare 1000 Play Isometric	Perfly Adult Badminton Racket BR Sensation 530
~18lb	~20lb	~22lb
R_1	R_2	R_3

For R_1:

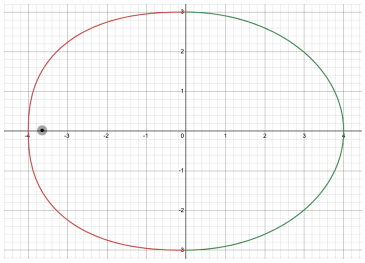
At $\delta = 2.476, L = 65.6\%$



At $\delta = 1.850, L = 72.2\%$

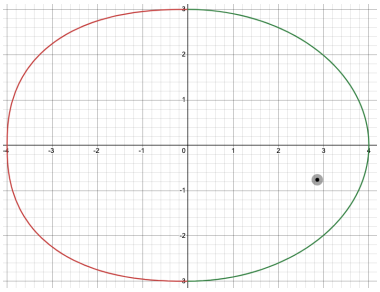


At $\delta = 2.587, L = 77.0\%$

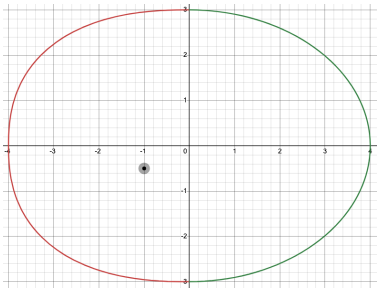


For R_2:

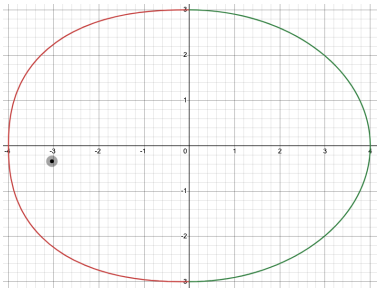
At $\delta = 2.528, L = 67.2\%$



At $\delta = 2.036, L = 74.2\%$

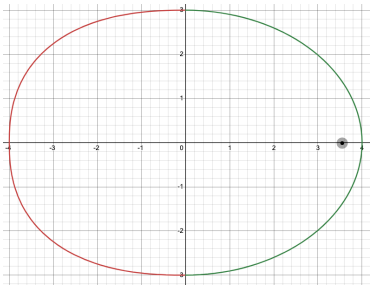


At $\delta = 2.636, L = 76.8\%$

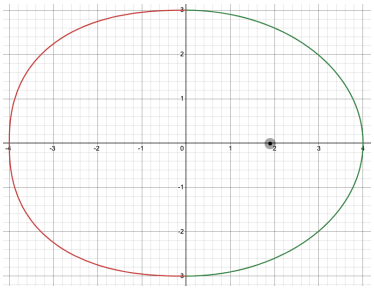


For R_3:

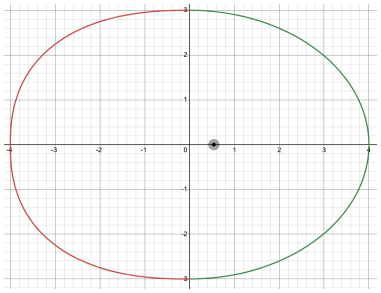
At $\delta = 2.466, L = 67.2\%$



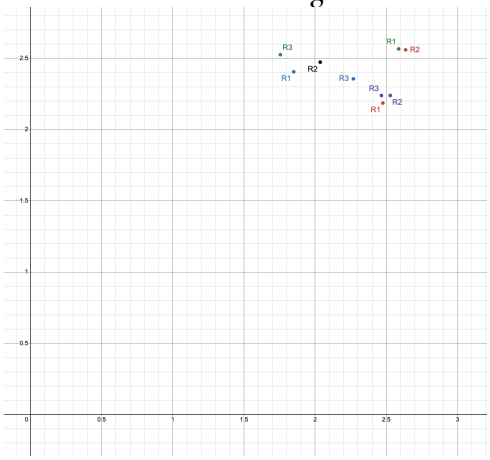
At $\delta = 2.269, L = 70.7\%$



At $\delta = 1.756, L = 75.8\%$



Distribution of Findings



[x-axis: δ, 1 unit = 1]
[y-axis: L, 1 unit = 30^{-1}]

Modelling a Curve

Creating a model of a curve by trial and error to fit each value would be a very tedious task. The findings were fed into ChatGPT and it was asked to approximate an equation fitting the values produced experimentally.

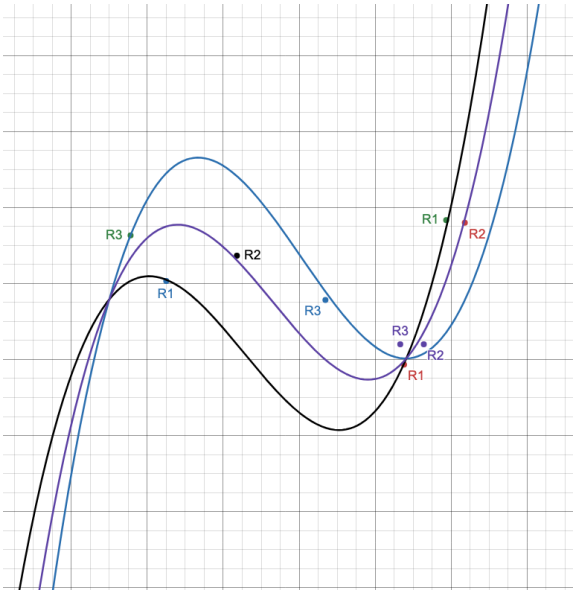
Let $L(\delta, T)$ be a multivariable function with δ representing impact location and T representing tension of the racket.

An empirical cubic approximation of $L(\delta, T)$ is:

$$L(\boldsymbol{\delta}, \boldsymbol{T}) = 65.2731 - 19.9433(\boldsymbol{\delta} - 2.3) + 98.5995(\boldsymbol{\delta} - 2.3)^2 + 194.4074(\boldsymbol{\delta} - 2.3)^3 + \frac{4.8387(\boldsymbol{T} - 20)}{2} - \frac{18.5292(\boldsymbol{\delta} - 2.3)(\boldsymbol{T} - 20)}{2} - \frac{44.3233(\boldsymbol{\delta} - 2.3)^2(\boldsymbol{T} - 20)}{2}$$

A mathematically equivalent but cleaner version would be:

$$L(\boldsymbol{\delta}, \boldsymbol{T}) = 194.4074\boldsymbol{\delta}^3 - 22.16165\boldsymbol{\delta}^2\boldsymbol{T} - 799.57856\boldsymbol{\delta}^2 + 92.67899\boldsymbol{\delta}\boldsymbol{T} + 758.164638\boldsymbol{\delta} - 93.5071985\boldsymbol{T} + 137.5231792$$



The black line represents $30L(x, 18)$
The purple line represents $30L(x, 20)$
The blue line represents $30L(x, 22)$
(A coefficient of 30 is given to each since the y-axis 1 unit = 1/30)

Keep in mind that this curve is a rough approximation, and the error may increase as δ varies or when T is high. This curve is useful in approximating the values of δ between ~1.6 and ~2.8.

For instance, to calculate the loss of energy of a shuttle that hits the racket at, say, 2.3 and the racket has a tension of 21lb, we'd do

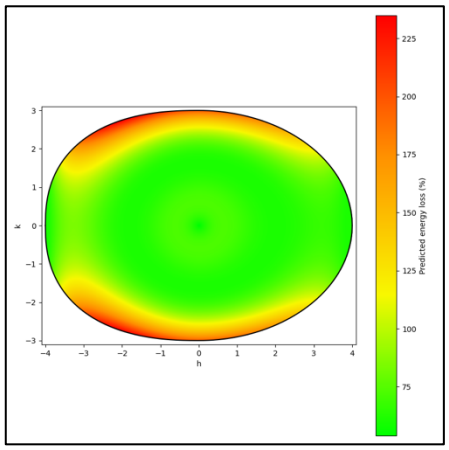
$$L(\mathbf{2.3}, \mathbf{21}) = 194.4074\mathbf{2.3^3} - 22.16165\mathbf{2.3^221} - 799.57856\mathbf{2.3^2} + 92.67899\mathbf{2.3 \times 21} + 758.164638\mathbf{2.3} - 93.5071985\mathbf{21} + 137.5231792 \approx 67.69\%$$

Hence a delta of 2.3 on a racket of tension 21lb yields approximately 67.69% loss of energy.

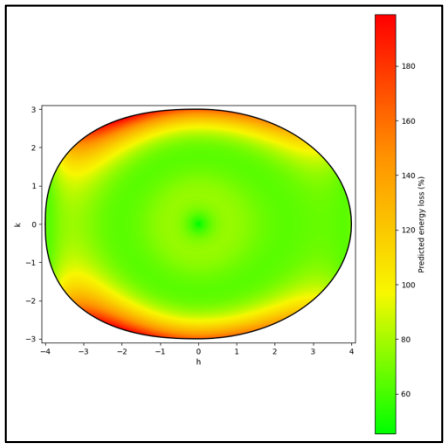
Creating a heatmap

It is possible to create a 'heatmap' of the racket showing the zones of higher and lower energy loss through Python. Using `matplotlib`. The colouring is based on the maximum and minimum outputs of the empirical energy loss function.

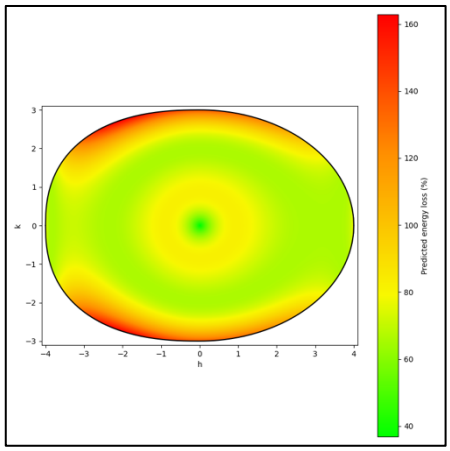
At T = 18:



At T = 20:



At T = 22:



Limitations

The limitations of the empirical model become most apparent near the edges of the racket, where the predicted energy-loss percentage increases sharply. This behaviour is primarily a consequence of the cubic nature of the fitted function, whose predictions can become increasingly sensitive to changes in the input variables, particularly outside the range of the experimental data.

The experimental setup was subject to several limitations, including the relatively small number of data points and the use of approximate measurements. These factors introduce uncertainty into the fitted function and may contribute to deviations between its predictions and actual collision behaviour. Nevertheless, the model provides a preliminary mathematical framework for investigating the relationship between impact location, racket tension and energy loss.

The model's reliability is expected to be greatest within the range of impact locations and tensions represented in the experimental data. At tensions or impact locations substantially removed from these observations, prediction errors may increase considerably. Consequently, the function should be interpreted as an empirical approximation rather than a universally applicable physical law. Further experimentation, with more data points, improved measurement precision and a broader range of controlled tensions, would help refine the model and assess its predictive validity.